

HOW TO SOLVE **HOMOGENEOUS** DIFFERENTIAL EQUATION

IN **7** EASY STEPS



$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

Step-by-Step Method
Made Simple!



Perfect for **IB AAHL** Students



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CONCEPTS MADE EASY



STEP-BY-STEP SOLUTIONS



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Syllabus Reference:



AHL 5.18

Content	Guidance, clarification and syllabus links
First order differential equations. Numerical solution of $\frac{dy}{dx} = f(x, y)$ using Euler's method.	$x_{n+1} = x_n + h$, where h is a constant.
Variables separable.	<i>Example:</i> the logistic equation $\frac{dn}{dt} = kn(a - n)$, $a, k \in \mathbb{R}$ Link to: partial fractions (AHL1.11) and use of partial fractions to rearrange the integrand (AHL5.15).
Homogeneous differential equation $\frac{dy}{dx} = f(\frac{y}{x})$ using the substitution $y = vx$.	

→ On 3/9/2026

Content	Guidance, clarification and syllabus links
Solution of $y' + P(x)y = Q(x)$, using the integrating factor.	

→ Tomorrow
10 p.m.



Question(1)–Paper–1

[Maximum mark: 13]

(a) Consider the differential equation

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right), \quad x > 0.$$

Use the substitution $v = \frac{y}{x}$ to show that the general solution of this differential equation is

$$\int \frac{dv}{f(v) - v} = \ln x + C. \quad [4]$$

(b) Hence, or otherwise, solve the differential equation

$$\frac{dy}{dx} = \frac{4x^2 + 5xy + y^2}{x^2}, \quad x > 0,$$

given that $y = 2$ when $x = 1$. Give your answer in the form $y = g(x)$. [9]

(a) Let

$$v = \frac{y}{x}$$

$$\therefore \frac{dv}{dx} = \frac{\frac{dy}{dx} \cdot x - y}{x^2} = \frac{x f\left(\frac{y}{x}\right) - \frac{y}{x}}{x^2}$$



$$\therefore \frac{dv}{dx} = \frac{\frac{dy}{dx} \cdot x - y(1)}{x^2} = \frac{xf\left(\frac{y}{x}\right) - \frac{y}{x}}{x^2} = \frac{f(v)}{x} - \frac{1}{x} \left(\frac{y}{x}\right)$$

$$\frac{dv}{dx} = \frac{f(v)}{x} - \frac{v}{x} = \frac{f(v) - v}{x}$$

$$\therefore \frac{dv}{f(v) - v} = \frac{dx}{x}$$

$$\therefore \int \frac{dv}{f(v) - v} = \int \frac{dx}{x}$$

$$\therefore \int \frac{dv}{f(v) - v} = \ln(x) + C \quad (\text{is general solution})$$



(b) Let $v = \frac{y}{x}$

$$\therefore \frac{dy}{dx} = \frac{4x^2 + 5xy + y^2}{x^2}, \quad x > 0,$$

$$\therefore \frac{dy}{dx} = 4 + 5\left(\frac{y}{x}\right) + \left(\frac{y}{x}\right)^2$$

$$\left(v = \frac{y}{x}\right) \left\{ \frac{dy}{dx} = f\left(\frac{y}{x}\right) = f(v) \right.$$

Then, $f(v) = 4 + 5v + v^2$

And using general solution from part (a)



$$\therefore \int \frac{dv}{f(v) - v} = \ln(x) + C$$

$$\therefore \int \frac{dv}{(4 + 5v + v^2) - v} = \ln(x) + c$$

$$\int \frac{dv}{(v + 2)^2} = \ln(x) + c$$

$$\int (v + 2)^{-2} dv = \ln(x) + c$$

$$\therefore \frac{(v + 2)^{-1}}{-1} = \ln(x) + c$$



$$\therefore -\frac{1}{2+v} = \ln(x) + c$$

Now, $y = 2$, when $x = 1$, and so $v = 2$, we get

$$\left(\because V = \frac{y}{x} = \frac{2}{1} = 2 \right)$$

$$c = -\frac{1}{4}$$

Substituting the the variable v , we obtain an expression for y

$$\ln x - \frac{1}{4} = -\frac{1}{2+v}$$

$$2+v = \frac{1}{\frac{1}{4} - \ln x}$$



$$\therefore 2 + \left(\frac{y}{x}\right) = \frac{1}{\frac{1}{4} - \ln x}$$

$$y = x \left(\frac{1}{\frac{1}{4} - \ln x} - 2 \right)$$

$$y = x \left(\frac{\frac{1}{2} + 2 \ln x}{\frac{1}{4} - \ln x} \right)$$

$$\begin{aligned} & \frac{1 - 2\left(\frac{1}{4} - \ln x\right)}{\frac{1}{4} - \ln x} \\ &= \frac{\left(1 - \frac{1}{2}\right) + 2 \ln x}{\frac{1}{4} - \ln x} \\ &= \frac{\frac{1}{2} + 2 \ln x}{\frac{1}{4} - \ln x} \end{aligned}$$



Practice questions PDF link is given in the description of this video.



HOW TO SOLVE A DIFFERENTIAL EQUATION

IN 5 EASY STEPS

USING INTEGRATING FACTOR

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$$\frac{dy}{dx} + P(x)y = Q(x)$$

5 EASY STEPS

- 1 Put in Standard Form
- 2 Find Integrating Factor (IF)
- 3 Multiply Throughout by IF
- 4 Integrate Both Sides
- 5 Solve for y

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